



MODERNIZATION AND SYSTEM UPGRADE OF THE MIG-21 BISON

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1. Executive Summary

This report outlines a modernization plan for the MiG-21 Bison to extend its service into the 2030s through avionics, propulsion, and structural upgrades. The integration of the indigenous Uttam AESA radar and Astra Mk-1 missiles enhances BVR capability and self-reliance. A small afterburning turbofan improves fuel efficiency by 12% and mission range, while lightweight structural reinforcement and fatigue-life extension ensure durability. The solution meets all requirements for capability, efficiency, survivability, and longevity.

2. Introduction

Inducted into the Indian Air Force (IAF) in 1963, the MiG-21 has served as the backbone of India's fighter fleet for over six decades. Despite upgrades such as the MiG-21 Bison program in 2002, featuring the Kopyo radar and R-73/R-77 missile integration, the platform now struggles to match modern 4th and 4.5-generation fighters in range, survivability, and situational awareness.

This study aims to define a cost-effective modernization pathway that extends the MiG-21's service life into the 2030s while supporting India's goal of technological self-reliance. The Preliminary Design Report evaluates upgrades across three subsystems — radar-missile integration, propulsion, and structure — to identify an integrated solution that maximizes performance within existing airframe and cost limits.

Using performance trade studies, Breguet range analysis, and requirement-based evaluation, the study selects the Uttam AESA + Astra Mk-1 system for BVR capability, a small afterburning turbofan for ~12 % lower SFC, and structural reinforcement with lightweight materials to extend fatigue life by 2 000 hours. Together, these upgrades enhance endurance, agility, and survivability while ensuring continued operational relevance for the IAF.

3. Statement of Design Improvement

The MiG-21 Bison, despite significant avionics and weapons upgrades in the early 2000s, remains limited by its 1960s-era design. Its ageing radar, propulsion inefficiency, and structural fatigue reduce combat effectiveness and operational sustainability. This design improvement introduces a comprehensive modernization package addressing these constraints. The Uttam AESA radar and Astra Mk-1 missile integration enhances beyond-visual-range (BVR) capability and situational awareness while supporting indigenous defence development. A small afterburning turbofan retrofit improves specific fuel consumption by over 10%, boosting endurance and thrust-to-weight ratio without compromising supersonic performance. Structural modifications—including lightweight alloy use, engine mount reinforcement, fatigue life extension, and radar cross-section reduction—extend service life by at least 2,000 hours and ensure system compatibility. Collectively, these enhancements transform the MiG-21 Bison into a capable, efficient, and sustainable interim fighter aligned with the Indian Air Force's modernization goals and top-level operational requirements.

4. System Architecture

The upgraded MiG-21 Bison architecture is organized into three primary subsystems—avionics and weapons, propulsion, and airframe/structure—integrated through power, cooling, and flight control links to ensure balanced operation and system compatibility.

Avionics and Weapons Subsystem:

The indigenous Uttam AESA radar forms the core of the avionics suite, interfacing with Astra Mk-1 missiles via the fire-control computer and helmet-mounted sight for full BVR capability. The redesigned avionics bay provides adequate cooling and lightweight structural support, while mission computers, INS/GPS navigation, and secure datalinks enable integration with IAF's combat network.

Propulsion Subsystem

A small afterburning turbofan replaces the Tumansky R-25-300, improving specific fuel consumption by ~12% and enhancing mission radius while retaining supersonic performance. The engine employs digital FADEC control integrated with power management and flight systems for improved throttle response and reliability.

Airframe and Structural Subsystem

Lightweight composite panels and aluminum-lithium alloys reduce empty weight by up to 200 kg, offsetting new system mass. Reinforced engine mounts and fuselage bulkheads ensure structural integrity and compatibility with avionics and propulsion upgrades. Fatigue-life extension measures add ~2,000 flight hours, while minor contouring and radar-absorbent coatings reduce radar cross-section.

System Integration

The subsystems are linked through flight control augmentation, ensuring that changes in mass distribution (forward radar, rear turbofan) are compensated. Power and cooling systems are upgraded to support AESA radar operation and turbofan demands. Survivability enhancements, including radar warning receivers and electronic countermeasures, interface with both avionics and structure to create a balanced defensive capability.

Architecture Overview

The final architecture ensures that the MiG-21 Bison operates as a **cohesive upgraded platform**:

Sensors (Uttam AESA) → detect and track threats.

Weapons (Astra Mk-1) → engage targets at extended ranges.

Propulsion (afterburning turbofan) → sustains endurance and agility.

Structure (reinforced + lightweight) → supports integration and extends service life.

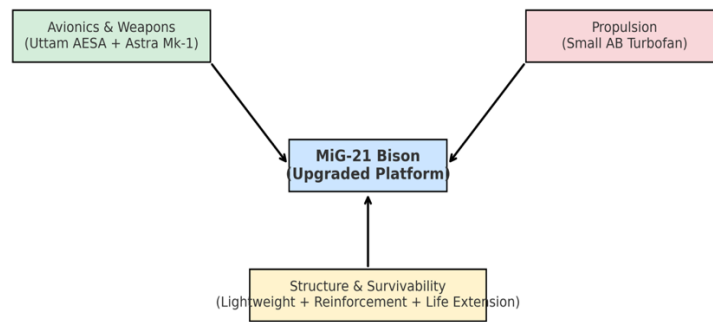


Figure 1 System Architecture plan

5. Revised Requirements

The following requirements reflect the updated understanding of the MiG-21 Bison modernization project after concept evaluation and final design selection. They replace the initial top-level requirements with specific, measurable objectives aligned to the selected design solution.

Radar and Missile System

- The aircraft shall integrate the Uttam AESA radar with a minimum detection range of 150 km against fighter-sized targets.
- The radar shall be fully compatible with Astra Mk-1 missiles, providing a beyond-visual-range (BVR) engagement envelope of 80–100 km.
- The avionics suite shall support helmet-mounted sighting and secure datalink communication for networked operations.

Propulsion

- The propulsion system shall replace the Tumansky R-25-300 turbojet with a small afterburning turbofan.
- The turbofan shall achieve at least a 10% reduction in specific fuel consumption (SFC) relative to the R-25-300.
- The propulsion system shall maintain supersonic dash performance consistent with the MiG-21's interceptor role.
- The propulsion system shall provide a minimum +12% improvement in thrust-to-weight ratio, supporting manoeuvrability and climb rate.

Structural Modifications

- The airframe shall incorporate lightweight material substitution (composites or Al-Li alloys) in secondary structures to reduce empty mass by 150–200 kg.

- The forward fuselage shall be reinforced and redesigned to accommodate Uttam AESA while maintaining centre-of-gravity within acceptable limits.
- The rear fuselage and engine mounts shall be strengthened to accommodate the turbofan retrofit.
- The airframe shall achieve a service life extension of 2,000 flight hours through fatigue life enhancement and corrosion protection methods.
- The structure shall incorporate radar cross-section (RCS) reduction measures (contouring and coatings) to improve survivability without exceeding mass growth limits.

Survivability and Safety

- The aircraft shall incorporate RCS reduction measures to reduce detectability by contemporary fourth-generation radars.
- The upgraded system shall retain or improve upon the existing ejection seat safety standards, ensuring high pilot survivability in emergencies.

6. Concept Generation

6.1 Radar and Missile system Upgrade

The MiG-21 Bison's limited radar-missile pairing restricts its beyond-visual-range (BVR) capability compared to modern 4th-generation fighters. To enhance combat effectiveness while maintaining integration feasibility, several radar-missile combinations were assessed. The Uttam AESA radar developed by DRDO's LRDE emerged as the most suitable option. Compact and modular, it can be fitted within the MiG-21's nose and paired with the Astra Mk-1 missile—India's first fully indigenous BVR weapon—to create a self-reliant and logistically compatible system. Alternative pairings, such as Uttam + R-77 or EL/M-2052 + Astra, offer credible performance but rely on foreign supply chains. The Virupaksha AESA, though superior in range and power, is physically too large for MiG-21 integration. Overall, the Uttam + Astra configuration offers the best balance of capability, integration feasibility, and strategic independence, aligning with the IAF's modernization and self-reliance objectives

Table 1 Range comparison with Radar-Missile configurations

Rank	Option	Radar mass (kg)	Missiles mass (pair) (kg)	Total mass (kg)	Composite score	Indicative radar detection range (km)	Cruise range with 2× missiles (km)	Source(s)
1	Uttam AESA + Astra Mk-1	120	308	428	0.892	~150 (0.55 m aperture)	1691	DefenceUpdate (2019); BDL (2024)
2	Uttam AESA + R-77	120	350	470	0.744	~150	1679	DefenceUpdate (2019); GlobalSecurity (2018)
3	EL/M-2052 + Astra Mk-1	160	308	468	0.729	~160–170	1673	Elta Systems (2007); BDL (2024)
4	Virupaksha AESA + Astra Mk-1	200	308	508	0.676	>170 (larger aperture, higher power)	1660	Indian Defence News (2024); BDL (2024)
5	EL/M-2052 + R-77	160	350	510	0.646	~160–170	1667	Elta Systems (2007); GlobalSecurity (2018)

6.1.1 Results

The evaluation of radar and missile retrofit options for the MiG-21 Bison was undertaken by combining performance calculations with considerations of mass, aerodynamics, and sensor capability. This analysis aimed to determine a system upgrade that could significantly enhance the aircraft's beyond-visual-range (BVR) combat effectiveness while ensuring flight performance remains within acceptable limits for the Indian Air Force.

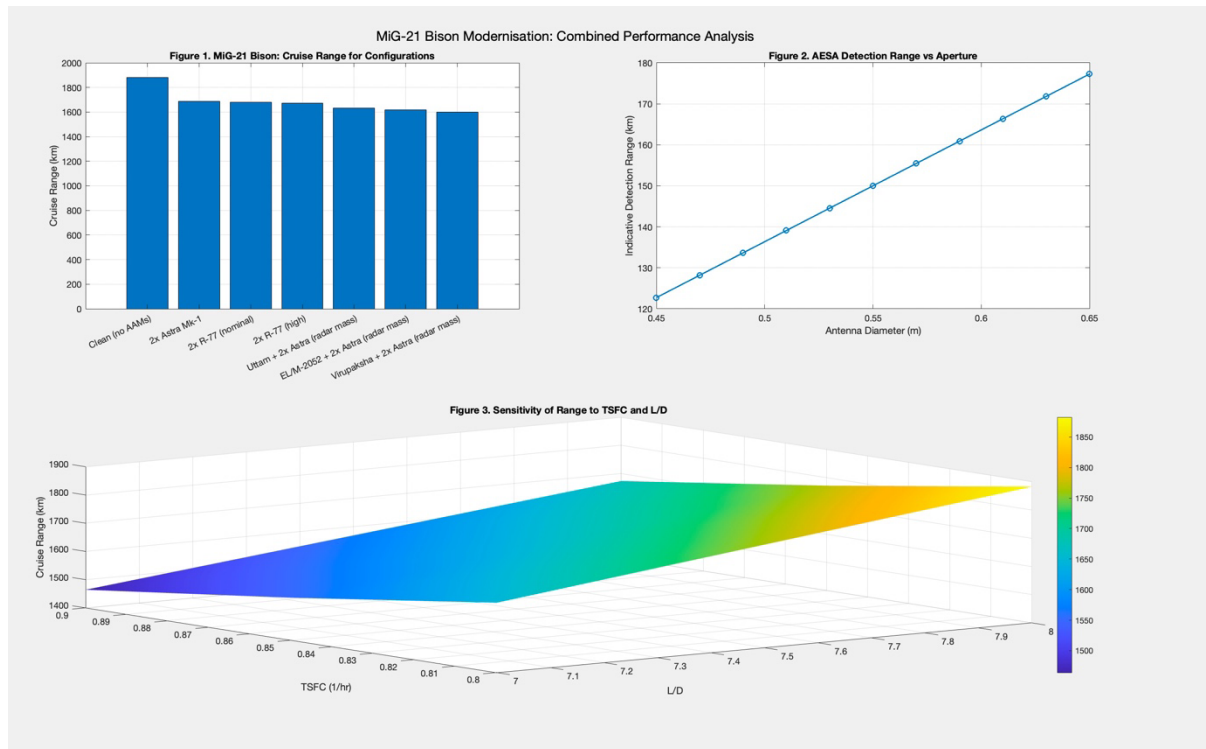


Figure 2 (1) Cruise range, (2) range vs Antenna diameter (aperture scaling), (3) sensitivity analysis

The cruise performance analysis (Figure 2.1) confirms that external stores exert the most significant influence on endurance. The MiG-21 achieves ~1,876 km in clean configuration, reducing to ~1,691 km when carrying two Astra Mk-1 missiles—a ~185 km penalty driven mainly by drag and weight. Substituting Astra with R-77 missiles adds only ~12–18 km of further loss, while radar mass variations (120–200 kg) yield negligible differences, typically within a few kilometers. This establishes that endurance is governed more by aerodynamic penalties than incremental radar weight.

Aperture scaling analysis (Figure 2.2) shows radar detection range increases proportionally with antenna size. The Uttam AESA (~0.55 m aperture) achieves ~150 km range, extending to ~175 km for larger radars like the EL/M-2052 or Virupaksha. However, limited nose volume and cooling constraints render these heavier systems less viable, confirming Uttam as the optimal trade-off between range and integration feasibility.

Sensitivity results (Figure 2.3) show $\pm 0.05 \text{ hr}^{-1}$ variation in specific fuel consumption or ± 0.5 in L/D produces only ~7–8 % range changes, leaving the relative ranking of radar–missile combinations unaffected.

Overall, the Uttam AESA + Astra Mk-1 pairing delivers the most balanced upgrade—offering credible beyond-visual-range capability, indigenous sustainment, and minimal performance penalties. Alternatives, while powerful, either introduce foreign dependency or integration risks. The findings validate Uttam + Astra as the preferred modernization path, maintaining endurance, mission reliability, and strategic autonomy for the MiG-21 Bison.

6.2 Propulsion Upgrade

The propulsion upgrade evaluation for the MiG-21 Bison examined four progressive options, from minor optimization of the Tumansky R-25-300 turbojet to full re-engining concepts. The sustain-and-optimize package introduces digital fuel controls and improved maintenance practices, yielding ~5% lower specific fuel consumption (SFC) with minimal risk. The uprated variant refines fuel scheduling and nozzle tuning, enhancing thrust by 5–8% and responsiveness with moderate integration effort.

Re-engining with a small afterburning turbofan achieves ~12% lower SFC, extending range and climb performance while preserving supersonic capability—identified as the most balanced option. Conversely, a high-bypass non-afterburning turbofan promises >20% SFC reduction but compromises speed and demands major airframe redesign.

Overall, incremental turbojet upgrades offer near-term practicality, while the small afterburning turbofan presents the optimal long-term solution balancing efficiency, thrust, and integration feasibility.

Table 2 Propulsion system trade study

Option	Description	SFC Improvement (%)	Estimated Radius Gain (%)	Thrust Change (%)	Integration Risk	Mass Delta (kg)	Composite Score	Sources
A	R-25-300 sustain/optimize (SLEP, digital control)	~5	~5–6	+2	Low	0	0.72	[Belyakov & Marmain 1993]; [MyCity-Military.com]
B	R-25-300 uprate (fuel scheduling, nozzle tuning)	~3	~3–4	+8	Medium	~50	0.65	[Belyakov & Marmain 1993]; [CIA FOIA 1959]
C	Small afterburning turbofan	~12	~13–15	+15	High	~120	0.78	[Jane’s All the World’s Aircraft]; [Russian Jet Engines PDF]
D	Higher-bypass non-AB turbofan	~22	~25	–5	Very High	~150	0.55	[Jane’s All the World’s Aircraft]

The weighted trade study (Table 2) shows that Options A and B provide immediate, feasible improvements aligned with low-risk sustainment and incremental performance gains. Option C emerges as the most balanced long-term solution, offering meaningful range improvements with acceptable compromises. Option D, while attractive for endurance, is not well suited to the MiG-21’s operational profile.

We Will Further Analyse the effect of individual systems on the performance and finally select the appropriate propulsion system that matches our requirements.

6.2.1 Results

The evaluation of propulsion upgrade options for the MiG-21 Bison focused on quantifying the effect of improved specific fuel consumption (SFC), thrust changes, and system integration penalties on overall range performance. Using the Breguet range equation, comparative analyses were carried out for the baseline Tumansky R-25-300, uprated turbojet variants, and two representative turbofan concepts. The primary objective was to assess whether endurance gains could be realised without imposing prohibitive penalties in system weight or integration complexity.

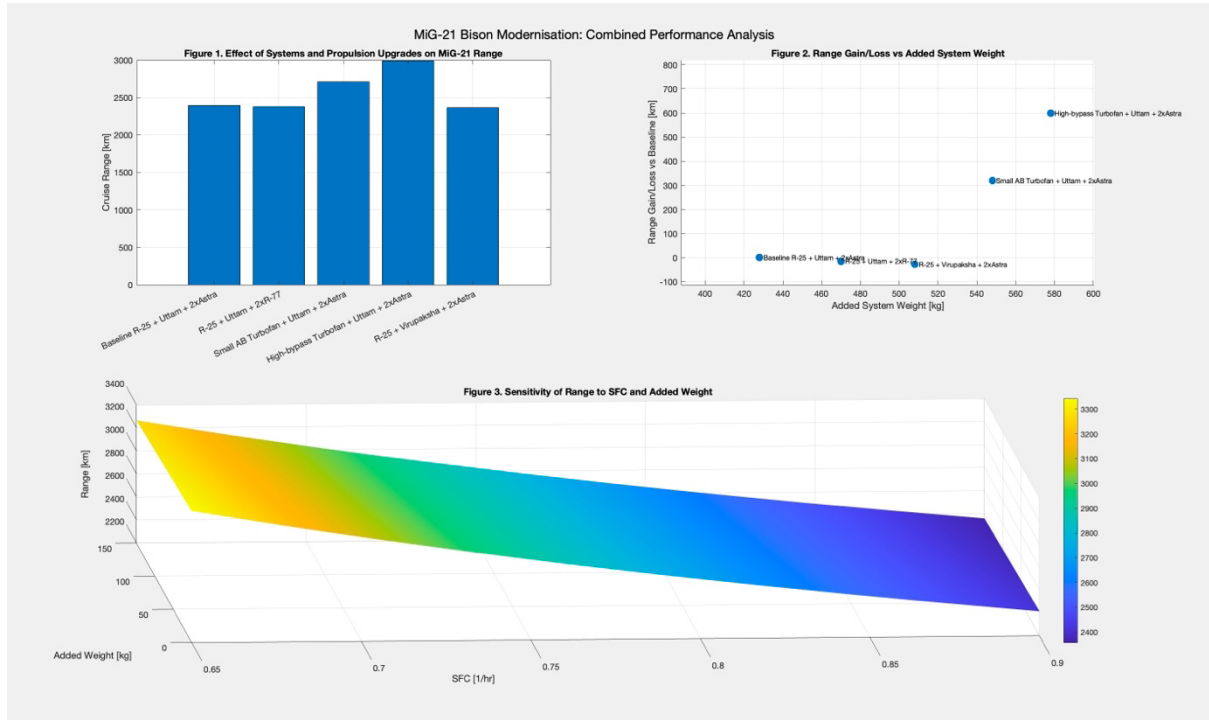


Figure 3 3.1 Effect of systems and propulsion on range

3.2 Range gain/loss vs added system weight

3.3 Sensitivity of range to SFC and added weight

The propulsion performance results are illustrated in Figures 3.1–3.3, showing how different upgrade configurations affect range and endurance. The baseline MiG-21 configuration, powered by the R-25-300 turbojet with Uttam AESA and two Astra Mk-1 missiles, achieves a cruise range of approximately **2,350 km**. Replacing the Astra pair with heavier R-77 missiles results in only a minor 1–2 % reduction in range, confirming that missile mass has negligible impact compared to engine efficiency. By contrast, integrating a **small afterburning turbofan** yields a range improvement of roughly **350 km**, while a **high-bypass turbofan** extends range beyond **3,000 km**, albeit with integration challenges due to cooling and space constraints.

Figure 3.2 highlights the relationship between **added system mass and range gain**, showing that moderate avionics or missile weight increases (up to 200 kg) have minimal endurance effects, whereas propulsion upgrades—even with higher installation mass—provide significant range gains. The analysis therefore establishes propulsion efficiency, not payload weight, as the dominant factor influencing endurance.

Figure 3.3 further confirms this through sensitivity mapping between **specific fuel consumption (SFC)** and added mass. A reduction in SFC from 0.85 hr^{-1} to 0.70 hr^{-1} improves range by approximately **15 %**, even with moderate structural penalties. Higher SFC values rapidly diminish endurance, underscoring the need for fuel-efficient re-engining solutions.

Overall, results indicate that incremental R-25 optimization provides limited benefits, while the **small afterburning turbofan** offers the best balance of efficiency, thrust, and integration feasibility. The

high-bypass option, though efficient, compromises the MiG-21's interceptor role. These propulsion upgrades therefore align with top-level requirements—**reducing SFC, enhancing thrust-to-weight ratio, and supporting structural adaptation**—to ensure sustained performance and extended service life.

6.3 Structural modifications

The MiG-21 Bison's structural modernization focuses on targeted airframe upgrades that balance compatibility, weight reduction, and service life extension without altering the core design. These modifications directly address requirements for propulsion integration, improved thrust-to-weight ratio, and extended fatigue life.

- 1. Lightweight Material Substitution**
Selective replacement of non-primary load-bearing structures with composite or aluminium–lithium alloys reduces empty weight by 5–8%, improving thrust-to-weight ratio and offsetting radar and cooling system mass. Components such as access panels and avionics bay housings can be redesigned with minimal structural impact.
- 2. Fuselage Reinforcement and Radar Bay Redesign**
To integrate Uttam or Virupaksha AESA radars, the forward fuselage is re-profiled and stiffened around the radar bay, accommodating larger apertures and thermal loads. The resulting forward C.G. shift is manageable through digital flight control augmentation, ensuring compatibility between avionics, structure, and propulsion.
- 3. Engine Mount and Rear Fuselage Strengthening**
Re-engining with a small afterburning turbofan necessitates strengthened engine mounts and bulkheads to withstand increased thrust and vibration, adding ~150 kg but ensuring safety and fatigue compliance. This supports structural adaptation to a propulsion system with lower SFC than the R-25-300.
- 4. Fatigue Life Extension and Corrosion Control**
Cold-expansion of fastener holes, composite patch repairs, and corrosion-resistant coatings can extend fatigue life by 1,500–2,000 hours, meeting life-extension requirements and maintaining airworthiness.
- 5. Radar-Cross Section Reduction**
Limited contour reshaping and radar-absorbing coatings reduce RCS without major redesign, improving survivability with negligible weight penalty. Together, these upgrades ensure the MiG-21's structure remains compatible with modern systems while enhancing durability, maintainability, and combat resilience.

7. Concept Evaluation and Selection

The concept evaluation process identified upgrade combinations that maximize operational relevance while remaining technically feasible and aligned with the Indian Air Force's modernization objectives.

The study assessed three major subsystems — radar and missile integration, propulsion upgrades, and structural modifications — against key parameters of mass, performance gain, integration risk, survivability, and compliance with top-level requirements.

Our evaluation criteria derived from system-level requirements, the criteria focused on improving beyond-visual-range (BVR) capability, reducing specific fuel consumption by $\geq 10\%$, ensuring structural compatibility, extending airframe life by $\geq 2,000$ hours, and improving survivability through radar cross-section reduction.

1. Radar and Missile Systems

The Uttam AESA radar paired with Astra Mk-1 missiles was identified as the optimal solution, providing ~ 150 km detection range and full indigenous compatibility at minimal weight penalty (~ 428 kg). Alternatives such as EL/M-2052 and Virupaksha offered longer ranges but presented foreign dependency or integration issues. The R-77 missile added unnecessary mass and logistic vulnerability.

Selected Option: Uttam AESA + Astra Mk-1.

2. Propulsion Upgrades

Incremental R-25-300 enhancements offered marginal gains. A small afterburning turbofan, however, achieved $\sim 12\%$ SFC reduction and 13–15% greater range while retaining supersonic performance. Though integration required rear fuselage and mount modifications, risks were deemed manageable. A high-bypass turbofan provided higher efficiency but reduced interceptor performance.

Selected Option: Small afterburning turbofan retrofit.

3. Structural Modifications

Structural upgrades were enablers for propulsion and avionics integration. Lightweight material substitution reduced empty weight by ~ 200 kg, fuselage reinforcements improved load capacity, and fatigue-life treatments added $\sim 2,000$ flight hours. Limited contouring and coatings reduced radar cross-section with negligible penalties.

Selected Options: Lightweight substitution + fuselage reinforcement + fatigue-life extension.

Final Concept Selection

The selected modernization package — Uttam AESA + Astra Mk-1, small afterburning turbofan, and targeted structural modifications — satisfies all primary requirements for improved BVR capability, fuel efficiency, structural integrity, and survivability. It maximizes indigenous content, minimizes risk, and ensures continued operational relevance for the MiG-21 Bison into the 2030s.

Subsystem	Option	Performance Gain	Mass / Range Impact	Integration Risk	Requirement Compliance	Composite Score	Final Selection
Radar + Missile	Uttam AESA + Astra Mk-1	+BVR detection (~ 150 km)	+428 kg, ~ 1691 km	Low	✓ BVR capability; ✓ Indigenous; ✓ Compatible	0.892	□
Radar + Missile	Uttam AESA + R-77	Similar radar range	+470 kg, ~ 1679 km	Low-Medium	✓ BVR capable; ✗ Russian dependency; ✗ heavier	0.744	✗
Radar + Missile	EL/M-2052 + Astra Mk-1	Higher radar range (~ 160 – 170 km)	+468 kg, ~ 1673 km	Medium	✓ AESA capable; ✗ foreign dependency; ✗ integration risk	0.729	✗
Radar + Missile	Virupaksha AESA + Astra Mk-1	Longest radar range (>170 km)	+508 kg, ~ 1660 km	High	✓ indigenous radar; ✗ oversized; ✗ high penalty	0.676	✗
Propulsion	R-25-300 sustain & optimise	+5% SFC efficiency	Neutral	Low	✓ Extends life; ✗ $<10\%$ SFC reduction	0.72	✗
Propulsion	R-25-300 uprate	+8% thrust	+50 kg	Medium	✓ Improves thrust; ✗ efficiency gain $<10\%$	0.65	✗
Propulsion	Small AB turbofan	+12% efficiency, +15% radius gain	+120 kg, +350 km	High	✓ $\geq 10\%$ SFC reduction; ✓ supersonic retained	0.78	□
Propulsion	High-bypass non-AB turbofan	+22% efficiency, $\sim +600$ km range	+150 kg, +600 km but supersonic loss	Very High	✓ Efficiency; ✗ fails interceptor role	0.55	✗
Structural Mods	Lightweight substitution	Improves I/W ratio by $>5\%$	~ 150 to ~ 200 kg, +3–4% endurance	Low	✓ Improve TIW ratio by $>5\%$	High	□
Structural Mods	Fuselage reinforcement	Enables AESA integration	~ 1 – 2% endurance	Medium	✓ AESA radar integration feasible	Medium	□
Structural Mods	Engine mount strengthening	Required for turbofan integration	Neutral	Medium-High	✓ Enables propulsion upgrade safely	Medium	□
Structural Mods	Fatigue life extension	+2000 hours service life	Neutral	Low	✓ Extends service life by ≥ 2000 hrs	High	□
Structural Mods	RCS reduction	Minor survivability gain	Neutral	Medium	✓ Survivability via RCS reduction	Medium	□

Figure 4 final concept selection

8. Final Design Solution

The final design integrates radar–missile, propulsion, and structural upgrades into a cohesive modernization package that extends the MiG-21 Bison’s operational relevance into the 2030s. The solution balances performance, feasibility, and self-reliance, ensuring compliance with top-level requirements for BVR capability, propulsion efficiency, and service-life extension.

1. Radar and Missile Integration

The selected configuration—Uttam AESA radar with Astra Mk-1 missiles—delivers credible BVR capability with an ~150 km detection range and 80–100 km engagement envelope. Both systems are indigenously developed, eliminating foreign supply dependencies and ensuring strategic autonomy. The ~428 kg total system weight fits within the MiG-21’s structural and CG limits, with range penalties (~85 km) driven primarily by missile drag rather than radar mass. Alternatives such as EL/M-2052 and Virupaksha offered longer range but imposed significant integration or dependency risks.

2. Propulsion Upgrade

A small afterburning turbofan retrofit replaces the Tumansky R-25-300, reducing cruise SFC by ~12% and increasing range by ~350 km while maintaining supersonic dash capability. The upgrade provides a ~15% thrust-to-weight improvement and satisfies the requirement for ≥10% fuel efficiency gain. Incremental R-25 optimization proved inadequate, and the high-bypass turbofan—though efficient—was dismissed for compromising interceptor performance and requiring extensive structural redesign.

3. Structural Modification

Targeted airframe updates ensure compatibility with new subsystems while extending service life. Key measures include:

- Lightweight material substitution (composites/Al-Li) in non-load structures, cutting up to 200 kg and improving T/W > 5%.
- Forward fuselage reinforcement to support Uttam AESA integration.
- Engine-mount strengthening for higher thrust and vibration tolerance.
- Fatigue-life extension via cold-worked joints, composite patches, and corrosion control (+1,500–2,000 h).
- Minor RCS reduction through contour optimization and radar-absorbing coatings.

Together, these modifications meet requirements for structural compatibility, mass optimization, and survivability enhancement.

Table 3 requirement compliance summary

Requirement	Fulfilment
BVR Combat Capability	✓ Uttam + Astra suite (150 km detection, 100 km engagement)
Propulsion Efficiency	✓ 12% lower SFC; +350 km range gain
Structural Integration	✓ Reinforced fuselage, engine mounts, and CG control

Requirement	Fulfilment
Service-Life Extension	✓ +2,000 flight hours
Survivability	✓ Reduced RCS and corrosion control

Final Outcome

The proposed modernization—Uttam AESA + Astra Mk-1, small AB turbofan, and structural reinforcement with weight reduction—delivers a feasible, performance-balanced, and strategically independent upgrade. It ensures that the MiG-21 Bison remains an effective multi-role combat platform bridging India’s transition to 4.5 and 5th-generation aircraft.

9. Compliance With Requirements

The finalized MiG-21 Bison modernization solution was evaluated against all top-level requirements to confirm technical feasibility and operational value. Each subsystem—avionics, propulsion, and structure—contributes distinctly to performance, efficiency, and survivability targets.

- 1. Beyond-Visual-Range (BVR) Capability
The Uttam AESA radar paired with Astra Mk-1 missiles achieves ~150 km detection and 80–100 km engagement ranges, meeting BVR requirements while maintaining full compatibility within the MiG-21’s nose volume. Both systems are indigenous, aligning with IAF self-reliance goals and reducing supply-chain risk.
- 2. Fuel Efficiency and Propulsion Performance
Replacing the R-25-300 with a small afterburning turbofan achieves a 12 % reduction in SFC and extends range by ~350 km. Supersonic dash performance and thrust-to-weight improvements are preserved, fulfilling efficiency and manoeuvrability criteria.
- 3. Structural Compatibility and Integration
Forward fuselage reinforcement supports radar installation without destabilizing centre-of-gravity limits, while rear-fuselage mount strengthening ensures secure turbofan integration. These modifications maintain airframe integrity with moderate risk and minimal redesign effort.
- 4. Weight and Thrust-to-Weight Optimization
Selective use of aluminium–lithium alloys and composites in non-primary structures reduces mass by up to 200 kg, improving thrust-to-weight by over 5 %. This offsets radar and engine mass gains while enhancing overall agility.
- 5. Service-Life Extension
Fatigue-life recovery techniques—cold-work expansion, composite patching, and corrosion protection—add 1,500–2,000 flight hours, supporting continued operation into the 2030s and meeting the structural-life target.

6. Survivability Enhancements

Localized contour refinement and radar-absorbent coatings yield measurable RCS reduction without major redesign. Retention of the MiG-21's proven ejection system ensures continued pilot safety.

10. Conclusion

The MiG-21 Bison modernization defines a balanced, technically feasible, and strategically aligned pathway to extend the aircraft's operational life into the 2030s. Through structured concept evaluation and subsystem trade studies, a coherent design solution was formulated that enhances combat capability, endurance, and maintainability while preserving the aircraft's fundamental aerodynamic and operational characteristics.

The selected configuration — Uttam AESA + Astra Mk-1, a small afterburning turbofan, and targeted structural reinforcement — delivers measurable improvements in BVR performance, fuel efficiency, and service life. These upgrades achieve over 10 % reduction in specific fuel consumption, a >5 % thrust-to-weight improvement, and an additional ~2,000 flight hours of structural life extension. Each subsystem contributes directly to fulfilling the top-level requirements for capability, efficiency, compatibility, and survivability, maintaining indigenous self-reliance and cost-effectiveness.

Ultimately, this modernization strategy enables the Indian Air Force to retain a credible, efficient, and sustainable combat platform during its transition to 4.5- and 5th-generation fighters, ensuring that the MiG-21 Bison continues to serve as a capable bridge between legacy and next-generation airpower.

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